

UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

Anonymous authors
Anonymous submission

ABSTRACT

Vision-language models (VLMs) have made substantial progress across a wide range of visual question answering benchmarks, spanning visual reasoning, document understanding, and multimodal dialogue. These improvements are evident across many VLMs built on a variety of base models, alignment architectures, and training data. Yet recent work shows that these same models trail on traditional image classification benchmarks, which probe fine-grained visual knowledge. We quantify this disconnect at scale: evaluating 15 recent VLMs on four fine-grained classification benchmarks, we find that models with near-identical general VQA scores can differ by as much as 19 percentage points (pp) in fine-grained accuracy, and that even the strongest VLM still lags its own CLIP vision encoder by 4.6pp. To pinpoint what closes this gap, we run 22 controlled ablations that vary one design choice at a time. The comparisons yield a consistent and interpretable pattern: upgrading the base LLM (Vicuna-7B→Qwen2-7B) lifts fine-grained and general VQA accuracy in lockstep (+7.5pp and +7.4pp), whereas a stronger vision encoder (CLIP ViT-L/14→DFN-CLIP ViT-H/14) selectively benefits fine-grained recognition (+4.5pp vs. +1.2pp on general VQA). Pretraining is likewise decisive for fine-grained performance—especially when the language model is unfrozen during pretraining, which alone adds +5.5pp—while pretraining-data quality matters comparatively little. Stacking these changes raises fine-grained accuracy from 52.8% to 73.4%, closing more than half of the gap to a state-of-the-art VLM. Together these measured margins chart a concrete path toward stronger fine-grained visual understanding and vision-centric capabilities in VLMs.

1 INTRODUCTION

Recent advances in vision-language models (VLMs), which integrate vision encoders with large language models (LLMs), have demonstrated increasingly sophisticated capabilities in interpreting and reasoning about visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). These models can perform complex tasks such as open-ended visual question answering, document understanding, and multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). Despite these advances, a crucial question remains largely unquantified: exactly how well do VLMs perform on fine-grained visual perception, how far do they fall short of dedicated vision encoders, and which design choices measurably close that gap? This paper answers these questions with a controlled, head-to-head measurement study.

Fine-grained visual recognition—the ability to distinguish between visually similar categories—requires models to focus on subtle distinguishing features while maintaining robustness to intra-class variations in pose, lighting, and background. Traditional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) set a high, well-documented bar on these benchmarks; by contrast, the fine-grained accuracy of newer VLMs has rarely been measured directly, and—as we show—these VLMs sit well below the encoders they are built on.

Understanding the fine-grained knowledge capabilities of VLMs is essential because perception serves as the foundation for advanced capabilities such as understanding and reasoning. Real-world applications of VLMs often involve fine-grained, long-tailed distributions where distinguishing between closely related categories is crucial. For instance, if a model fails to identify a mushroom species correctly, it cannot determine whether it is poisonous based on textual knowledge or

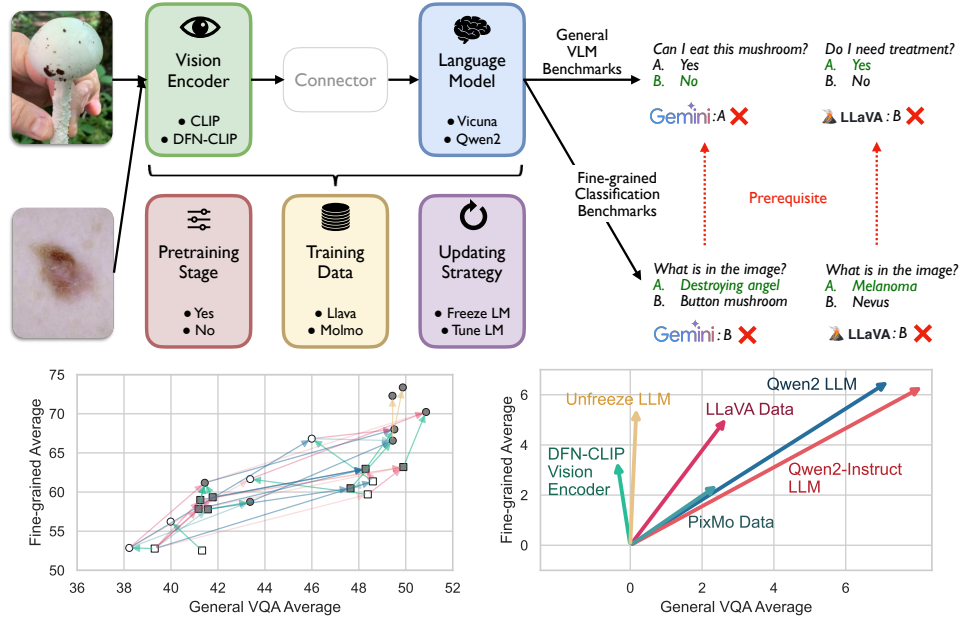


Figure 1: **Overview.** (Top) We investigate the fine-grained classification capabilities of vision-language models (VLMs), a crucial yet often overlooked aspect that underpins higher-level understanding and reasoning. (Bottom) Across 22 controlled ablations that each isolate one model component or training strategy (left), the measured accuracy deltas clearly separate the factors that drive fine-grained classification from those that do not (right), quantifying the contribution of every design choice.

whether it is safe for consumption. Quantifying *which* components govern this capability—and by how much—therefore directly informs more effective architectures and training strategies for future VLMs, which is precisely what our controlled comparisons deliver.

In this paper, we first present a comprehensive, apples-to-apples evaluation of fine-grained visual knowledge in state-of-the-art VLMs, placing their fine-grained classification accuracy side by side with their general VLM performance. Specifically, we evaluate 15 VLMs—including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024)—on four fine-grained classification benchmarks (ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014)) against aggregated results from eight general VLM benchmarks such as MMMU (Yue et al., 2024), Math-Vista (Lu et al., 2024), and MMVet (Yu et al., 2024). The comparison is striking: models with essentially identical general VQA scores diverge by up to 19pp in fine-grained accuracy, demonstrating that fine-grained recognition is a distinct axis of visual intelligence that existing VLM benchmarks simply do not measure. We further quantify a persistent shortfall between every VLM and its own CLIP-based vision encoder—consistent with, and now precisely measured against, prior observations (Zhang et al., 2024; Geigle et al., 2024).

To pinpoint what drives fine-grained classification performance in VLMs, we conduct a systematic study that varies one design choice at a time and reads off the resulting accuracy delta. Specifically, we measure how VLM components (language model, vision encoder) and training strategies (pretraining stage, weight-updating method, data quality) each move fine-grained versus general VQA accuracy. Across 22 carefully controlled ablation experiments, the measured margins isolate several factors that significantly—and, in some cases, selectively—affect fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** A stronger base LLM lifts both axes in near-equal measure—swapping Vicuna-7B for Qwen2-7B adds +7.5pp fine-grained and +7.4pp general VQA accuracy—so it improves fine-grained recognition but does not *close the gap* to it.
- **Vision encoder (§4.2):** A stronger vision encoder acts selectively: moving from CLIP ViT-L/14 to DFN-CLIP ViT-H/14 yields +4.5pp on fine-grained classification but only +1.2pp on general VQA (in properly pretrained, two-stage models).

- **Pretraining stage (§4.3):** Large-scale image-caption pretraining is disproportionately valuable for fine-grained accuracy (e.g., +6.1pp for Vicuna-based VLMs) while moving general VQA far less (+3.1pp).
- **Weight updating method (§4.3):** Unfreezing the LLM alongside the connector during pretraining—rather than tuning the connector alone—adds a further +5.5pp on fine-grained classification with essentially no cost to general VQA.
- **Data quality (§4.3):** Caption quality barely moves fine-grained accuracy: substituting richly annotated PixMo captions for noisy web-scraped LLaVA captions shifts fine-grained performance by only −1.6pp, particularly when the language model is frozen during pre-training.

In summary, our study measures fine-grained visual perception in VLMs—an essential yet overlooked aspect of current VLM benchmark evaluations—and, through controlled ablations, attaches a concrete accuracy figure to each contributing factor. Cumulatively, these interventions raise fine-grained accuracy from 52.8% to 73.4% (and general VQA from 39.3% to 49.9%), closing more than half of the gap to a state-of-the-art VLM and offering an empirically grounded recipe for building vision-centric VLMs that excel on real-world long-tail tasks demanding fine-grained recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) are a class of models that integrate visual inputs with language models, typically following the LLaVA-based architecture (Liu et al., 2023) of three core components: a vision encoder, a language model, and an MLP connector bridging the two. These models have achieved strong performance across various multimodal tasks, including visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). To evaluate VLMs, numerous benchmarks have been developed, such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). However, these benchmarks primarily assess reasoning and language understanding given visual inputs, while overlooking core vision-centric capabilities such as object recognition and fine-grained classification. To bridge this gap, we systematically *measure* VLMs on fine-grained classification and, crucially, contrast those measurements against general VQA scores—an essential yet underexplored axis of visual intelligence on which, as our numbers show, otherwise-comparable models diverge sharply.

Fine-grained visual classification. Fine-grained classification is a well-established computer vision task that aims to distinguish between visually similar subcategories within broader object categories, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). These benchmarks have played a key role in training and evaluating vision encoders, like ResNet (He et al., 2016), CLIP (Radford et al., 2021) and DINO (Caron et al., 2021). Despite their importance in vision research, recent VLMs have largely overlooked these datasets in their evaluation protocols in favor of visual question-answering. Since fine-grained classification is critical for real-world applications that involve long-tail distributions and require precise recognition (e.g., medical diagnosis, food safety, and species identification), evaluating VLMs on such benchmarks is essential for understanding their practical usability from a vision-centric perspective, beyond language-centric reasoning tasks.

Evaluation and ablation of VLMs for fine-grained classification. While recent studies have explored VLM evaluation on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), they have been limited in scope, often focusing on just a few models or datasets. We broaden the comparison to 15 VLMs across four fine-grained classification benchmarks under a single, unified multiple-choice protocol, making cross-model numbers directly comparable. Beyond this wider measurement, we contribute 22 controlled ablations that assign a quantitative contribution to each design choice—turning qualitative observations into head-to-head accuracy margins.

Several recent works have investigated the design space of VLMs, analyzing aspects such as model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the role of multimodal datasets (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). However, these studies primarily report gains on general, language- and reasoning-focused benchmarks. In this work, we quantify how each key design choice—base model selection, pretraining method,

and the composition of pretraining and instruction-tuning data—separately shifts fine-grained versus general accuracy, providing measured, vision-centric optimization targets rather than aggregate improvements alone.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

In this section, we motivate the evaluation of fine-grained classification, describe our benchmarks and evaluation setup, and report the quantitative patterns that emerge from testing 15 existing VLMs side by side against both general VQA benchmarks and their own CLIP vision encoders.

3.1 MOTIVATION

Fine-grained classification is a fundamental task in computer vision that focuses on distinguishing visually similar subcategories within a broader category (e.g., different bird species or car models) (Wei et al., 2021). Many evaluation benchmarks, such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008), have been developed and widely used as standard testbeds for training and evaluating vision encoders. With the advent of large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) achieved near- or superhuman performance on challenging fine-grained classification tasks.

However, these benchmarks have been overlooked in the evaluation of vision-language models (VLMs). While recent VLMs like LLaVA (Liu et al., 2023) demonstrate strong capabilities in interpreting visual concepts, their reported numbers come almost exclusively from visual question answering (VQA)—academic benchmarks like MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or document understanding tasks like DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). Because these benchmarks emphasize reasoning and language processing and contain few fine-grained object categories, a model’s vision-centric accuracy can be high on paper yet remain entirely unmeasured—exactly the blind spot our comparisons expose.

Fine-grained classification is crucial for VLMs, as real-world applications often encounter long-tailed distributions that require precise visual knowledge between similar categories. For example, in daily life, if a blind person asks whether a mushroom is safe to eat, the model must first correctly identify the species before reasoning about its toxicity—misclassifying a deadly destroying angel as an edible white button mushroom could have fatal consequences. Similarly, in medical diagnosis, failing to distinguish between similar diseases could lead to incorrect treatment and serious health risks. In self-driving cars, confusing a “stop” sign with a “do not enter” sign—both red with similar shapes—could result in dangerous navigation errors. Therefore, it is critical for VLMs to excel at fine-grained classification to ensure safety and reliability in real-world scenarios.

3.2 BENCHMARKS

We evaluate the fine-grained visual knowledge of VLMs using four well-established object recognition benchmarks, each focusing on different domains of fine-grained classification: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

These benchmarks span 37 to 1,000 distinct classes, making direct evaluation of VLMs in a multiple-choice format—commonly used in VLM training and evaluation—challenging. To address this while keeping the comparison fair, we adopt the methodology of Geigle et al. (2024) to convert each dataset into 5-way multiple-choice questions. Specifically, for each image we use an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) to mine the four hardest negative choices by image-label cosine similarity, then combine them with the ground-truth label. We deliberately use OpenCLIP for negative mining so the difficulty is not biased against the VLMs under test, which rely on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023)—all trained on datasets distinct from OpenCLIP’s. Critically, this conversion lets us test VLMs in their native format while retaining most of the task difficulty: on the 102-class Flowers-102, CLIP ViT-L/14-336px (Radford et al., 2021) scores 75.2% under full open-set classification and only 78.3% on our 5-way version—a mere 3.1pp gap that confirms the multiple-choice framing preserves, rather than trivializes, the underlying challenge.

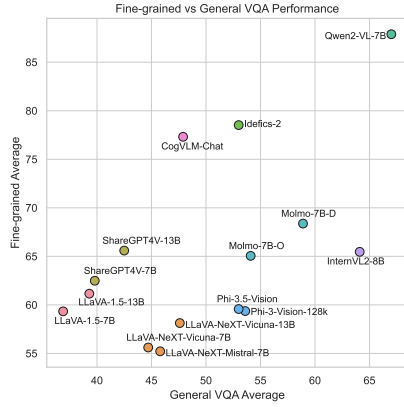


Figure 2: **Fine-grained classification vs. general VQA accuracy across 15 VLMs.** Each point is one VLM; the wide vertical spread at any fixed general-VQA score—up to 19pp between models with matched VQA performance—shows that fine-grained classification is a distinct axis of visual capability that standard VQA benchmarks do not measure.

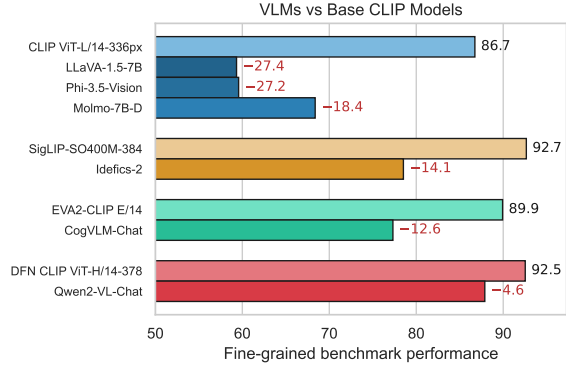


Figure 3: **Every VLM trails its own CLIP vision encoder on fine-grained classification.** Bars pair each VLM with the encoder it was built on. Qwen2-VL-Chat comes closest, still 4.6pp behind DFN-CLIP, while the remaining VLMs fall much further short (e.g., Molmo at 68.4% vs. its CLIP ViT-L/14 at 86.7%, an 18.3pp gap)—direct evidence that substantial fine-grained knowledge present in the encoder is lost in the VLM.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abdin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b), holding the parameter budget roughly fixed so that measured differences reflect design rather than scale. For each VLM, we use its default prompt format from VLMEvalKit (Duan et al., 2024) and report accuracy as the exact match with the multiple-choice options. For baseline comparison, we evaluate various CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) in the standard zero-shot setup, where both images and class labels (“A photo of a [class]”) are encoded using CLIP’s image and text encoders. The top-1 prediction is determined by computing the cosine similarity between image and text embeddings.

To test whether existing VLM benchmarks already capture fine-grained ability, we place each model’s fine-grained classification accuracy directly against its average over eight general VQA benchmarks, such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), from VLMEvalKit—the paired comparison that drives Finding 1.

3.4 RESULTS

Figure 2 plots fine-grained classification accuracy (averaged over our four benchmarks) against general VQA accuracy (averaged over eight existing VQA benchmarks) for all 15 tested VLMs. The two axes decouple sharply: at any given level of general VQA performance, fine-grained accuracy varies widely across models. The clearest case is a matched pair—CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) sit within a fraction of a point of each other on general VQA (both $\approx 48\%$), yet their fine-grained accuracy differs by a full 19pp (77.3% vs. 58.1%). A 19pp swing at fixed general-VQA performance is a large effect, and it establishes directly that fine-grained classification is not captured by existing VLM benchmarks: it is a separate, measurable dimension of visual intelligence.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Because every current VLM is built on a pretrained vision encoder, the multiple-choice format lets us compare each VLM head-to-head against the very encoder it inherits—an unusually clean baseline, since the encoder’s fine-grained knowledge is already inside the model. Figure 3 makes this comparison, and the gap is consistent and one-directional: *every* VLM underperforms its own encoder. The gap is largest for Molmo (Deitke et al., 2024), a very recent model that excels on general VQA benchmarks yet reaches only 68.4% average accuracy here—18.3pp below its own CLIP ViT-L/14 (Radford et al., 2021) encoder at 86.7%. Even the best VLM we test, Qwen2-VL-Chat (Wang et al., 2024b), still trails its state-of-the-art DFN-CLIP (Fang et al., 2023) encoder by 4.6pp—meaning the strongest available integration still leaves measurable fine-grained knowledge on the table. That the shortfall holds across every model, from best to worst, motivates a controlled search for the design choices that recover this lost accuracy, which we take up in the next section.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on the observational gaps in Section 3, we now isolate the specific design choices behind them through controlled ablations, varying one factor at a time so that each measured accuracy delta is attributable to a single change. To keep every comparison clean, all runs share the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023), one epoch of pretraining (when applicable) followed by one epoch of finetuning, and LLaVA-1.5’s default hyperparameters (including learning rate and batch size). Holding the pipeline fixed in this way is what lets us read a change in fine-grained accuracy as the effect of the ablated component rather than of incidental training differences.

We report fine-grained performance as the average over the four classification benchmarks of Section 3.2, and general VQA performance as the average over three widely used multiple-choice benchmarks—MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b)—chosen for their domain coverage and widespread adoption. Throughout, we track both axes together so that a design choice’s *selectivity* (fine-grained vs. general) is visible, not just its overall effect.

4.2 BASE MODELS

Section 3 showed that recent models typically pair a newer base LLM with the same CLIP ViT-L/14 vision encoder from Radford et al. (2021). We quantify how much each of these two base-model choices moves fine-grained accuracy—and whether it moves general VQA accuracy by the same amount.

4.2.1 LLM CHOICE

Setup. Using LLaVA-1.5-7B (Liu et al., 2023) as our baseline, we ablate the LLM component across different vision encoders and pretraining configurations. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023) (used in LLaVA) against Qwen2-7B (Yang et al., 2024) (used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024)), as well as Llama-2-7B (Touvron et al., 2023) (Vicuna’s base model) and Qwen2-7B-Instruct (the instruction-tuned variant of Qwen2-7B).

Findings. As Figure 4 shows, replacing Vicuna-7B with Qwen2-7B improves performance in every configuration we tested, and the average gains are almost identical on the two axes: +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA. The instruction-tuned Qwen2-7B-Instruct reproduces this pattern (+7.5pp and +8.1pp). Llama2-7B, the non-instruction-tuned base of Vicuna, yields only modest gains, though the downstream implications of skipping instruction tuning remain unclear. Crucially, this near-parity is not an artifact of formatting: all trained models—including the lower-scoring Vicuna-based VLMs—produce correctly formatted multiple-choice responses. The measured gains therefore reflect genuinely improved knowledge about the

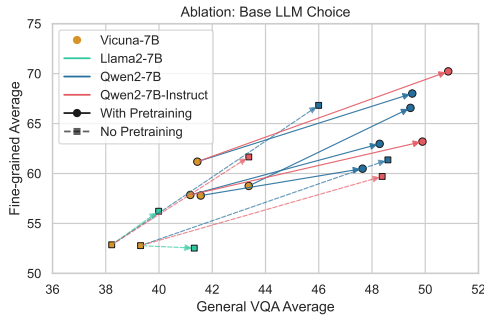


Figure 4: **Base-LLM ablation: Vicuna-7B vs. stronger LLMs, all else fixed.** The effect is large and, tellingly, near-symmetric across the two axes: replacing Vicuna with Qwen2-7B adds +7.5pp fine-grained and +7.7pp general VQA accuracy. A stronger LLM raises *both* kinds of performance in equal measure, so it cannot by itself account for the fine-grained-specific gap.

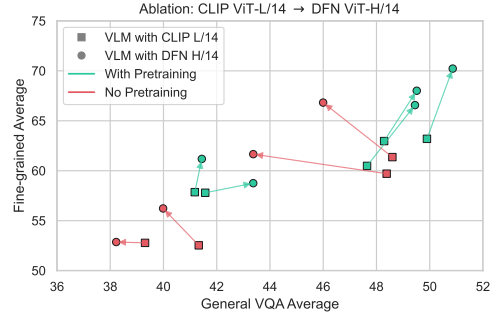


Figure 5: **Vision-encoder ablation: CLIP ViT-L/14 vs. DFN-CLIP ViT-H/14, all else fixed.** In contrast to the LLM, the encoder acts selectively: given sufficient pretraining, DFN-CLIP adds +4.5pp on fine-grained classification but only +1.2pp on general VQA—a nearly 4× larger effect on the fine-grained axis.

options rather than better format adherence, and their symmetry across axes is precisely why a stronger LLM improves fine-grained accuracy without narrowing the fine-grained-specific gap.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA series and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021), whereas Qwen2-VL (Wang et al., 2024b) uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which posts stronger zero-shot object-recognition scores on benchmarks like ImageNet. We swap between these two encoders across several LLM choices and pre-training configurations.

Findings. Figure 5 shows that the payoff from CLIP L/14 → DFN-CLIP H/14 hinges on whether connector pretraining precedes finetuning. Without pretraining, the stronger encoder is not yet usable: general VQA drops −2.5pp and fine-grained gains only +2.8pp. Once connector pretraining is in place, the picture flips decisively in favor of fine-grained recognition—general VQA holds steady or edges up (+1.2pp) while fine-grained accuracy jumps +4.5pp. The selectivity is the key quantitative signal here: a better encoder delivers most of its benefit to fine-grained classification, and only when properly integrated through pretraining.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pre-training before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are typically trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, trains on large-scale image-caption datasets via a captioning task to align the vision encoder and LLM. The second stage, instruction tuning, refines the VLM on curated multi-round, multi-modal instruction data to align it with human intent. We now quantify how much each stage—and each knob within it (data source, weight-updating method)—contributes to fine-grained versus general accuracy, so their relative importance can be compared directly.

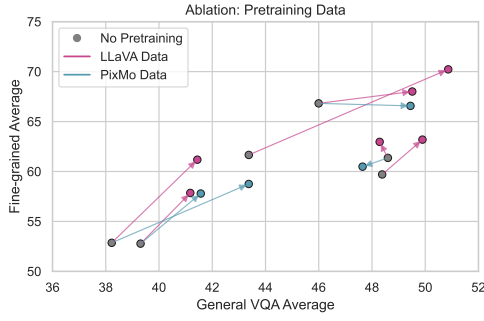


Figure 6: **Pretraining-data ablation: no pretraining vs. adding LLaVA (CC-3M) or Molmo (PixMo-Cap) data.** Both data sources help, and by similar amounts despite very different caption quality: LLaVA adds +4.0pp fine-grained / +2.1pp general, PixMo +2.4pp / +2.5pp. In both cases the fine-grained gain meets or exceeds the general-VQA gain, marking pretraining as a fine-grained-leaning intervention.

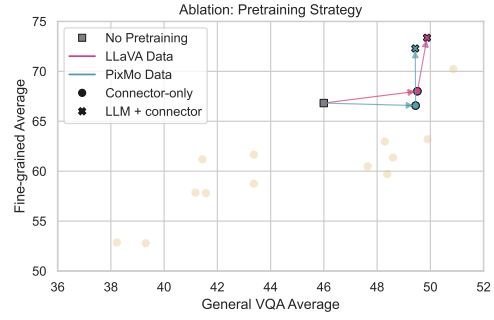


Figure 7: **Training-method ablation: connector-only pretraining vs. jointly tuning the LLM and connector.** Unfreezing the LLM during pretraining is a near-free fine-grained win: +5.5pp on fine-grained classification with no measurable cost to general VQA. The gain appears under both LLaVA and PixMo data, so it is a property of the training method rather than the caption source.

Table 1: **Key ablation settings with per-step accuracy deltas, alongside existing models (bolded).** Parenthesized values are the change from the row above. Reading down the block, upgrading the LLM (+5.1pp fine-grained), the vision encoder (+5.0pp), and jointly pretraining the connector and LLM (+5.4pp) each add substantial fine-grained accuracy, cumulatively moving 52.8%→73.4%; instruction finetuning contributes the least (+2.4pp). General VQA follows a different profile—dominated by the LLM swap (+7.1pp)—making the fine-grained-vs.-general contrast explicit column by column.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

4.3.1 PRETRAINING

Setup. We investigate several critical questions about the pretraining process:

- Is pretraining necessary for LLaVA-architecture models when the base models are already pretrained on billion-scale data?
- How does connector-only pretraining compare with unfreezing the LLM?
- How important is the quality of pretraining data for downstream performance?

We experiment with three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, with high-quality human-annotated detailed captions), and no pretraining. For most experiments, we pretrain only the connector following Liu et al. (2023). Additionally, we explore a strategy similar to Deitke et al. (2024), where we pretrain the connector for 20% of the steps before unfreezing the LLM for the remaining 80%. All configurations are followed by one epoch of finetuning on LLaVA finetuning data.

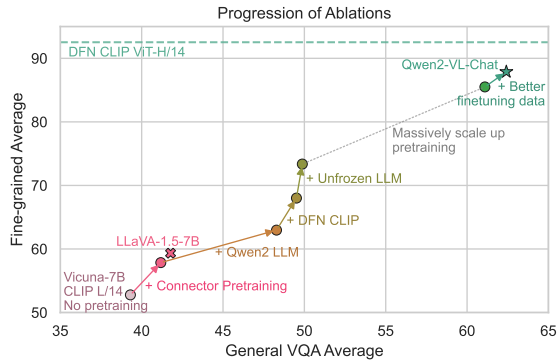


Figure 8: **Cumulative effect of stacking the ablated changes.** Starting from Vicuna-7B with CLIP ViT-L/14 at 52.8%, sequentially upgrading the LLM, vision encoder, and training settings lifts fine-grained accuracy to 73.4%—a +20.6pp climb that recovers more than half of the 35.1pp gap to Qwen2-VL-Chat’s 87.9%. The residual difference tracks the one factor we could not match: Qwen2-VL’s 1.4T-token pretraining corpus, orders of magnitude larger than ours.

Findings. Figure 6 reports the pretraining ablation. Connector-only pretraining improves nearly every configuration—the lone exception being the Qwen2 with CLIP ViT-L/14 pairing—and the improvement is again fine-grained-leaning. The effect is largest for Vicuna-based VLMs, which gain +6.1pp on fine-grained classification against +3.1pp on general VQA: pretraining buys about twice as much fine-grained accuracy as general-VQA accuracy for these models.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Strikingly, caption quality barely registers. Despite the two sources differing enormously in richness—web-scraped LLaVA captions versus detailed human PixMo annotations—switching from LLaVA to PixMo shifts fine-grained accuracy by only −1.6pp and general VQA by +0.4pp, both within the noise of the far larger effects reported above.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

One explanation for this insensitivity is that connector-only pretraining prevents the model from exploiting the richer captions at all. To test it, we pretrain both the LLM and connector, using a 20% connector-only warmup. Figure 7 shows this unlocks a further +5.5pp on fine-grained classification with no drop in general VQA—a clean, one-sided improvement. Yet the +5.5pp gain appears equally under LLaVA and PixMo data, so even when the LLM *can* use the captions, the higher-quality source confers no additional fine-grained benefit and the noisier captions do no harm to the LLM’s language ability.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. Departing from our earlier setup, we start from the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b) (the base of Qwen2-VL-7B-Chat) and finetune on LLaVA-Instruct data (Liu et al., 2023) under our protocol, then compare against Qwen2-VL-7B-Chat, whose finetuning uses 2M in-house examples (Wang et al., 2024b). Holding the base model fixed isolates the effect of the finetuning dataset alone.

Findings. Swapping Qwen2-VL’s 2M-example instruction set for LLaVA-Instruct data changes accuracy remarkably little—and slightly for the worse on both axes. Fine-grained accuracy falls just 2.4pp (87.9% → 85.5%) and general VQA 1.3pp (62.4% → 61.1%). Placed against the other interventions in Figure 8, this 2.4pp swing is the *smallest* fine-grained movement of any factor we

study—direct evidence that, at this stage, instruction finetuning is far less decisive for fine-grained knowledge than base-model choice or pretraining strategy.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 bring the ablations together, attributing a concrete accuracy contribution to each component as we walk from the weakest to a near-strongest configuration on both axes. Starting from the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) with CLIP ViT-L/14 (Radford et al., 2021)), the stacked changes raise fine-grained accuracy by +20.6pp (52.8% $\rightarrow$ 73.4%) and general VQA by +10.6pp (39.3% $\rightarrow$ 49.9%). The decomposition is clean and complementary: the DFN-CLIP (Fang et al., 2023) upgrade and unfreezing the LLM during pretraining deliver most of the *fine-grained* gain (a combined roughly +10pp) while barely touching general VQA, whereas the Qwen2-7B (Yang et al., 2024) LLM swap supplies most of the *general-VQA* gain—each design choice contributing to exactly the axis our earlier ablations predicted.

Even after stacking every change, about 12pp of fine-grained accuracy still separates our best configuration from Qwen2-VL-Chat (Table 1). We weigh two candidate explanations: architectural differences and pretraining-data scale.

The evidence points squarely at scale. Our observational results (Section 3.4) and training ablations (Section 4.3) already showed pretraining to be the dominant fine-grained lever, and the data budgets differ by orders of magnitude: our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) runs each used under 1M image-caption pairs ($\approx$ 200M and 400M tokens), whereas Wang et al. (2024b) pretrain Qwen2-VL on 1.4T tokens—several thousand times more. A gap of this magnitude on the very lever we found most fine-grained-sensitive is the most parsimonious account of the residual 12pp, and it underscores that large-scale pretraining is the key remaining ingredient for state-of-the-art fine-grained classification.

5 LIMITATIONS & CONCLUSION

Our study has practical limitations that open opportunities for future research. Under our compute budget, every ablation trained on <1 M image-caption pairs rather than the billion-scale corpora used by some newer VLMs; since pretraining scale emerged as our most fine-grained-sensitive factor—and the leading explanation for the residual ≈ 12 pp gap to Qwen2-VL-Chat—quantifying how these deltas extend to 1.4T-token regimes is the natural next step. Newer pretraining strategies may also interact differently with fine-grained understanding, and measuring their contributions in the same controlled framework would sharpen the picture further.

In this work, we turn a qualitative concern into measured evidence: across 15 VLMs, fine-grained accuracy varies by up to 19pp at matched general-VQA performance and every model trails its own vision encoder (by as much as 18.3pp), establishing fine-grained classification as a crucial, separately measurable dimension of VLM capability. Our 22 controlled ablations then attach a concrete margin to each design choice—a stronger LLM (+7.5pp, symmetric across axes), a stronger vision encoder (+4.5pp, fine-grained-selective), and joint LLM+connector pretraining (+5.5pp)—which together lift fine-grained accuracy from 52.8% to 73.4%. These quantified, axis-resolved findings offer an empirically grounded recipe for building vision-centric VLMs that perform reliably in real-world scenarios demanding precise visual understanding.

REFERENCES

- Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Jianmin Bao, Harkirat Behl, et al. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024. [2](#), [5](#)
- Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. In *NeurIPS*, 2022. [1](#)
- Anas Awadalla, Irena Gao, Josh Gardner, Jack Hessel, Yusuf Hanafy, Wanrong Zhu, Kalyani Marathe, Yonatan Bitton, Samir Gadre, Shiori Sagawa, et al. Openflamingo: An open-source framework for training large autoregressive vision-language models. *arXiv preprint arXiv:2308.01390*, 2023. [3](#)
- Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool. Food-101 – mining discriminative components with random forests. In *European Conference on Computer Vision*, 2014. [2](#), [4](#), [15](#)
- Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 9650–9660, 2021. [1](#), [3](#), [4](#)
- Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Conghui He, Jiaqi Wang, Feng Zhao, and Dahua Lin. Sharegpt4v: Improving large multi-modal models with better captions. In *European Conference on Computer Vision*, pp. 370–387. Springer, 2024a. [5](#)
- Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, et al. Are we on the right way for evaluating large vision-language models? In *NeurIPS*, 2024b. [6](#), [16](#)
- Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, et al. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 24185–24198, 2024c. [5](#), [7](#)
- Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E Gonzalez, et al. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality. *Technical Report*, 2023. [6](#), [10](#)
- Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning. In *NeurIPS*, 2023. [3](#)
- Matt Deitke, Christopher Clark, Sangho Lee, Rohun Tripathi, Yue Yang, Jae Sung Park, Mohammadreza Salehi, Niklas Muennighoff, Kyle Lo, Luca Soldaini, et al. Molmo and pixmo: Open weights and open data for state-of-the-art multimodal models. *arXiv preprint arXiv:2409.17146*, 2024. [2](#), [3](#), [5](#), [6](#), [8](#), [10](#), [17](#)
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *CVPR*, 2009. [2](#), [4](#), [15](#)
- Haodong Duan, Junming Yang, Yuxuan Qiao, Xinyu Fang, Lin Chen, Yuan Liu, Xiaoyi Dong, Yuhang Zang, Pan Zhang, Jiaqi Wang, et al. Vlmevalkit: An open-source toolkit for evaluating large multi-modality models. In *Proceedings of the 32nd ACM international conference on multimedia*, pp. 11198–11201, 2024. [5](#), [16](#)
- Alex Fang, Albin Madappally Jose, Amit Jain, Ludwig Schmidt, Alexander Toshev, and Vaishaal Shankar. Data filtering networks. *arXiv preprint arXiv:2309.17425*, 2023. [4](#), [5](#), [6](#), [7](#), [10](#)
- Samir Yitzhak Gadre, Gabriel Ilharco, Alex Fang, Jonathan Hayase, Georgios Smyrnis, Thao Nguyen, Ryan Marten, Mitchell Wortsman, Dhruva Ghosh, Jieyu Zhang, et al. Datacomp: In search of the next generation of multimodal datasets. In *NeurIPS*, 2023. [3](#)
- Gregor Geigle, Radu Timofte, and Goran Glavaš. African or european swallow? benchmarking large vision-language models for fine-grained object classification. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 2653–2669, 2024. [2](#), [3](#), [4](#), [15](#)
- Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *CVPR*, 2017. [1](#)

- Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, Dinesh Manocha, and Tianyi Zhou. Hallusionbench: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14375–14385, June 2024. 16
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016. 3
- Wenyi Hong, Weihan Wang, Ming Ding, Wenmeng Yu, Qingsong Lv, Yan Wang, Yean Cheng, Shiyu Huang, Junhui Ji, Zhao Xue, et al. Cogvlm2: Visual language models for image and video understanding. *arXiv preprint arXiv:2408.16500*, 2024. 5
- Siddharth Karamcheti, Suraj Nair, Ashwin Balakrishna, Percy Liang, Thomas Kollar, and Dorsa Sadigh. Prismatic vlms: Investigating the design space of visually-conditioned language models. *arXiv preprint arXiv:2402.07865*, 2024. 3
- Aniruddha Kembhavi, Mike Salvato, Eric Kolve, Minjoon Seo, Hannaneh Hajishirzi, and Ali Farhadi. A diagram is worth a dozen images. In *ECCV*, 2016. 16
- Hugo Laurençon, Léo Tronchon, Matthieu Cord, and Victor Sanh. What matters when building vision-language models? *arXiv preprint arXiv:2405.02246*, 2024. 5
- Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *NeurIPS*, 2023. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 16
- Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge, January 2024a. URL <https://llava-vl.github.io/blog/2024-01-30-llava-next/>. 2, 5
- Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. Mmbench: Is your multi-modal model an all-around player? In *European conference on computer vision*, pp. 216–233. Springer, 2024b. 6, 16
- Yuliang Liu, Zhang Li, Mingxin Huang, Biao Yang, Wenwen Yu, Chunyuan Li, Xu-Cheng Yin, Cheng-Lin Liu, Lianwen Jin, and Xiang Bai. Ocrbench: on the hidden mystery of ocr in large multimodal models. *Science China Information Sciences*, 67(12), December 2024c. ISSN 1869-1919. doi: 10.1007/s11432-024-4235-6. URL <http://dx.doi.org/10.1007/s11432-024-4235-6>. 16
- Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. In *ICLR*, 2024. 2, 4, 5, 16
- Minesh Mathew, Dimosthenis Karatzas, and CV Jawahar. Docvqa: A dataset for vqa on document images. In *WACV*, 2021. 1, 3, 4
- Brandon McKinzie, Zhe Gan, Jean-Philippe Fauconnier, Sam Dodge, Bowen Zhang, Philipp Dufter, Dhruvi Shah, Xianzhi Du, Futang Peng, Floris Weers, et al. Mml1: Methods, analysis & insights from multimodal llm pre-training. *arXiv preprint arXiv:2403.09611*, 2024. 3
- M-E. Nilsback and A. Zisserman. Automated flower classification over a large number of classes. In *Proceedings of the Indian Conference on Computer Vision, Graphics and Image Processing*, 2008. 2, 3, 4, 15
- OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023. 1
- O. M. Parkhi, A. Vedaldi, A. Zisserman, and C. V. Jawahar. Cats and dogs. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2012. 2, 3, 4, 15
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021. 1, 3, 4, 5, 6, 7, 10
- Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do imagenet classifiers generalize to imagenet? In *ICML*, 2019. 15

- Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade W Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, Patrick Schramowski, Srivatsa R Kundurthy, Katherine Crowson, Ludwig Schmidt, Robert Kaczmarczyk, and Jenia Jitsev. LAION-5b: An open large-scale dataset for training next generation image-text models. In *Thirty-sixth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2022. URL <https://openreview.net/forum?id=M3Y74vmsMcY>. 4, 5, 15
- Amanpreet Singh, Vivek Natarjan, Meet Shah, Yu Jiang, Xinlei Chen, Devi Parikh, and Marcus Rohrbach. Towards vqa models that can read. In *CVPR*, 2019. 4
- Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023. 1
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023. 6
- Vishaal Udandara, Ameya Prabhu, Adhiraj Ghosh, Yash Sharma, Philip HS Torr, Adel Bibi, Samuel Albanie, and Matthias Bethge. No” zero-shot” without exponential data: Pretraining concept frequency determines multimodal model performance. *arXiv preprint arXiv:2404.04125*, 2024. 3
- C. Wah, S. Branson, P. Welinder, P. Perona, and S. Belongie. Caltech-ucsd birds 200. Technical report, 2011. 3
- Ke Wang, Juntong Pan, Weikang Shi, Zimu Lu, Mingjie Zhan, and Hongsheng Li. Measuring multimodal mathematical reasoning with math-vision dataset. *arXiv preprint arXiv:2402.14804*, 2024a. 3
- Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024b. 1, 2, 3, 5, 6, 7, 9, 10
- Xiu-Shen Wei, Yi-Zhe Song, Oisín Mac Aodha, Jianxin Wu, Yuxin Peng, Jinhui Tang, Jian Yang, and Serge Belongie. Fine-grained image analysis with deep learning: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 44(12):8927–8948, 2021. 4
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2. 5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. 6, 10
- Weihao Yu, Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Xinchao Wang, and Lijuan Wang. Mm-vet: Evaluating large multimodal models for integrated capabilities. In *ICML*, 2024. 2, 16
- Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *CVPR*, 2024. 1, 2, 3, 4, 5, 6, 16
- Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 11975–11986, 2023. 4
- Yuhui Zhang, Alyssa Unell, Xiaohan Wang, Dhruva Ghosh, Yuchang Su, Ludwig Schmidt, and Serena Yeung-Levy. Why are visually-grounded language models bad at image classification? In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. 2, 3

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Full per-configuration ablation results (aggregate).** Each row is one controlled configuration; the two rightmost columns report averaged fine-grained Classification and general VLM accuracy, so the effect of any single design change can be read by comparing rows that differ in exactly one column.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMStar	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Per-configuration ablation results broken out by general VQA benchmark** (MMBench, MMMU, MMStar). Reporting each benchmark separately shows that the trends in the aggregate column are consistent across benchmarks rather than driven by any single one.

A FULL RESULTS

We present the complete ablation results for aggregated performance, general VQA benchmarks, and fine-grained classification in Table 2, Table 3, and Table 4, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Per-configuration ablation results broken out by fine-grained benchmark** (ImageNet, Flowers, Pets, Food). The per-dataset columns show that the vision-encoder and pretraining gains reported in the main text hold across all four domains, confirming the effects are broad rather than dataset-specific.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As covered in Section 3.2, we evaluate VLMs and base vision encoders on four different traditional classification benchmarks:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset covering a broad range of supercategories, each containing multiple fine-grained subcategories, based on the WordNet hierarchy. The ImageNet test set contains 50,000 images with classification across 1,000 classes.

However, the ImageNet/WordNet hierarchy contains many classes too visually distinct to yield hard multiple-choice distractors (for instance, only three classes fall under the “person” category). To address this, and the dataset’s excessive size, we use human-annotated incorrect model predictions from Recht et al. (2019) to select images for which genuinely hard negatives can be constructed. This filtering takes the pool from 19,056 annotated examples down to 5,457 images spanning 928 of the original 1,000 classes—retaining broad class coverage while ensuring every question is difficult. Because the resulting class distribution is uneven, we report mean-per-class accuracy for ImageNet so that frequent classes do not dominate the reported number.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A dataset of 102 flower species characterized by high intra-class variation and inter-class similarity. The Flowers dataset consists of 6,149 test examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A collection of 37 pet categories with challenging variations in pose, lighting, and occlusion. The Pets dataset consists of 3,669 test examples.

Food-101 (Bossard et al., 2014): A dataset comprising 101 food categories, exhibiting substantial visual diversity within each class. The Food dataset consists of 25,250 test examples.

Question generation. For all fine-grained classification test sets except the ImageNet dataset (described above), we generate hard negatives by performing zero-shot classification on the test examples using an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle

et al. (2024). We take the top 4 predictions that are incorrect, and shuffle them with the correct label to produce the list of answer choices.

Prompt formatting. In our multiple choice VQA format, the question is always of the form “What type of object is in this photo?” The answer choices are formatted as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. However, since different VLMs are trained with different multiple choice prompts and format, we use each model’s corresponding format from VLMEvalKit (Duan et al., 2024), for instance: “Answer with the option’s letter from the given choices directly.”

For CLIP models, we use the standard zero-shot evaluation procedure, and thus there is no question format. The classnames passed to the text encoder are instead formatted in the template of “a photo of a {classname}”.

B.2 GENERAL VQA

For the observational comparison in Section 3, we take general-VQA scores directly from the Open-VLM Leaderboard (Duan et al., 2024), whose default average spans eight diverse benchmarks: MM-Bench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This eight-benchmark average is the general-VQA axis used in Figure 2.

For all subsequent controlled evaluations—of both existing models and our own trained models in Section 4—we standardize on three multiple-choice benchmarks, giving a tighter, more directly comparable general-VQA measurement:

MMBench (Liu et al., 2024b): A dataset of 2,948 examples testing a wide range of VLM capabilities split between perception and reasoning. While the majority of the examples deal with coarse-grained perception and visual reasoning tasks, some examples test “fine-grained perception”, though the categories (such as action recognition, attribute recognition, and OCR) are in a different domain from the fine-grained classification datasets we consider.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset covering a wide distribution of domains and applications. Though the test set has 10,500 examples, most works cite model performance on the validation split, which contains 900 questions. MMMU tests general knowledge tied to visual perception across art, science, engineering, medicine, business, and humanities.

MMStar (Chen et al., 2024b): A dataset comprised of 1,500 questions selected by human annotators. The examples are chosen to cover a diverse set of tasks, both across domain and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024). All existing models have a pre-defined evaluation procedure, and for our own trained models, we apply the LLaVA evaluation code, since we exclusively finetune on LLaVA-Instruct data.

C TRAINING

We build on the LLaVA-1.5 training codebase (Liu et al., 2023). For the majority of experiments, we stick to the LLaVA architecture and training procedure.

Pretraining. This means one epoch of pretraining on the LLaVA pretraining data, obtained from CC-3M images and captions, or Molmo pretraining data (PixMo-Cap). For PixMo-Cap, we only use the “caption” field and not the raw audio transcript. During pretraining, we use a batch size of 256. In most of our pretraining runs, we tune only the randomly-initialized MLP connector with a learning rate of 1e-3. When ablating pretraining strategy, we first tune the connector only as warmup, for 20% of steps, then unfreeze the LLM and tune it as well, with a learning rate of 2e-5 for the remaining 80% of steps.

Finetuning. For finetuning, we always train for one epoch on the LLaVA-1.5 finetuning data, which consists of LLaVA-Instruct and VQA training data from various datasets to allow proper bench-

marking. During finetuning, we maintain a learning rate of $2e-5$ for the LLM, with a batch size of 128. The vision encoder remains frozen during both phases of training.

For the ablation experiment on the Qwen2-VL architecture, we start with Qwen2-VL-7B Base and train with the same method, unfreezing only the LLM.

Compute. We train all models using 4 nodes of 4 A100 (40GB) GPUs each. Training typically completes within 20 hours, with variation depending on the CLIP model size.

LLaVA vs. PixMo data. The main text contrasts the caption quality of the LLaVA and PixMo pre-training data. LLaVA pretraining data comes from CC-3M web-scraped captions, widely regarded as low in quality and relevance. PixMo-Cap, by contrast, is built from human annotators’ 60–90 second spoken descriptions, later consolidated into detailed captions (Deitke et al., 2024). The quality gap is stark by any surface measure: LLaVA captions average 9.8 words while PixMo captions average 169 words—a roughly $17\times$ difference in length. That such a large disparity in caption richness produces only a -1.6pp change in fine-grained accuracy is exactly what makes our data-quality finding notable.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”